

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

Approximating intractable posterior distributions efficiently is a central problem in Bayesian inference. Stein Variational Gradient Descent (SVGD) has become a widely adopted variational inference method, transporting a set of interacting particles so that their empirical distribution approximates a target measure. Because the SVGD dynamics unfold within a reproducing kernel Hilbert space (RKHS), their behavior is governed by the choice of kernel, which jointly determines the rate of convergence and the fidelity of the resulting approximation. The ubiquitous median heuristic furnishes a convenient rule for setting a single scalar bandwidth, but it is inflexible, lacks theoretical justification, and degrades markedly as the dimension of the state space grows. We address this limitation with a principled strategy for selecting kernel parameters adaptively over an arbitrary parameterized family of kernels. Recent convergence analyses phrased in terms of the kernelized Stein discrepancy (KSD) establish that maximizing the KSD over the kernel parameters accelerates the decrease of the Kullback–Leibler divergence; building directly on this principle, we introduce Adaptive SVGD (Ad-SVG), which interleaves each SVGD transport step with a gradient-ascent update of the kernel bandwidths on the KSD. We develop a self-contained analysis that lifts existing fixed-kernel convergence guarantees to the adaptive setting, proving that the maximal KSD over the kernel class vanishes along the iterates under both exact and approximate inner maximization and, under an additional Stein log-Sobolev condition, obtaining convergence in Kullback–Leibler divergence together with explicit iteration-complexity bounds. These guarantees are borne out empirically. Across a range of inference tasks—Gaussian mixtures, Gaussian targets of growing dimension, an ODE-based linear inverse problem, Gaussian-process inference, and Bayesian logistic regression—Ad-SVG matches the median heuristic wherever that heuristic already succeeds and decisively surpasses it as the dimension grows, correcting in particular the systematic variance underestimation that characterizes median-bandwidth SVGD.

1 Introduction

Stein Variational Gradient Descent (SVGD) [SVGD] is a deterministic, particle-based method for approximate Bayesian inference that has established itself as a compelling alternative to traditional Markov Chain Monte Carlo (MCMC) sampling. SVGD evolves an ensemble of interacting particles along the update directions obtained from the functional gradient of the Kullback–Leibler (KL) divergence to the target distribution, with the admissible directions constrained to the unit ball of a reproducing kernel Hilbert space (RKHS). This RKHS structure is at once the source of the method’s tractability and its principal vulnerability: the performance of SVGD is dictated by the choice of kernel, which shapes the interaction between particles and thereby the algorithm’s convergence [Duncan2019OnTG, Nsken2021SteinVG]. A further symptom of a poorly matched kernel is that the resulting particle approximation systematically underestimates the posterior variance [DBLP:conf/iclr/BaEGSSWZ22]. Taken together, these observations have hardened into

the widely held view that SVGD is intrinsically ill-suited to inference in higher dimensions. We take issue with this diagnosis. The difficulty commonly attributed to dimension is, to a substantial extent, a difficulty of kernel selection: by tuning the kernel adaptively during inference through maximization of the kernelized Stein discrepancy (KSD), the very same SVGD dynamics transport particles effectively across the moderate-dimensional and anisotropic targets on which the standard heuristic fails.

1.1 Related work.

Since its introduction [SVGD], SVGD has become a standard tool for approximate Bayesian inference across a broad spectrum of machine learning applications [DBLP:conf/uai/LiuRLP17, DBLP:conf/iclr/MessaoudMXP0CC24, DBLP:conf/nips/PuGHLHC17, DBLP:journals/tsp/Kassabov19] and its theoretical foundations have been the subject of sustained progress. Mean-field convergence has been characterized in both the continuous-time [LLN2019, Duncan2019OnTG, NEURIPS2020'16f8e136] and discrete-time [NEURIPS20203202111c, pmlr-v162-salim22a] regimes, and finite-particle convergence rates have been established under a variety of assumptions [balasubramanian2024improvedfiniteparticleconvergencerates, NEURIPS2023'54e5d7af]. Complementing these analyses, the SVGD dynamics have been identified with a gradient flow on the space of probability measures [DBLP:conf/nips/Liu17, Duncan2019OnTG], in direct analogy with the variational formulation of the Fokker–Planck equation of VariationalFormulationFokker.

The kernel is not a peripheral hyperparameter but the very object that determines the particle interactions, and hence the convergence, of the method (see also Figures 1 and 9). This dependence is mirrored in the theory: convergence results are typically stated as mean-field convergence with respect to the KSD, and whether such convergence in turn entails weak convergence is itself contingent on the chosen kernel [pmlr-v70-gorham17a]. Against this backdrop, the widely used median heuristic [JMLR:v13:gretton12a] recommends itself chiefly by its simplicity: it enjoys no theoretical guarantee and is known to deteriorate as the dimensionality of the problem grows [DBLP:conf/iclr/BaEGSSWZ22, DBLP:conf/icml/ZhuoLSZCZ18, DBLP:conf/icml/WangZ018]. A substantial body of work has consequently sought to counteract this high-dimensional degradation through structural modifications of the dynamics—including Stein Newton and second-order schemes, matrix-valued and preconditioned kernels, and sliced or projected variants [stein'newton, NEURIPS2019'5dcd0ddd, gong2021sliced, pmlr-v151-liu22a]—and, along a distinct line, through neural-network parameterizations that learn the update direction directly [langosco2022neural, ZHAO2023118975]. Our work is orthogonal and complementary to these efforts: rather than altering the transport dynamics, we intervene solely on the kernel, and the adaptive mechanism we propose could in principle be layered on top of many of these variants.

The approach most closely related to ours is that of ai'stein'2023, who introduce a mixture-of-kernels framework: they define a KSD for a weighted linear combination of a fixed set of base kernels and learn the mixture weights by maximizing this multiple-kernel KSD. Their construction is confined to a finite kernel basis and optimizes only over the simplex of mixture weights. We remove both restrictions, performing continuous optimization over the parameters of an arbitrary kernel family; the finite-mixture case is thereby recovered as a special instance of our general formulation.

1.2 Contributions.

This work is organized around a single, deliberately sharp research question: can SVGD, stripped of any architectural or algorithmic augmentation, attain competitive performance in moderate dimensions once it is equipped with a principled, adaptively chosen kernel? By isolating kernel selection from the many other modifications proposed for SVGD variants, we quantify the extent to which kernel choice alone governs the method’s effectiveness, and we establish how far adaptive kernel learning can carry the original algorithm. Our answer is affirmative: adaptive kernel selection recovers the accuracy that fixed heuristics forfeit, and it does so without altering the underlying particle dynamics.

Concretely, the paper makes three contributions.

- (i) **An adaptive kernel-selection method for SVGD.** We introduce Ad-SVG, which updates the kernel parameters during inference by ascending the KSD. Where the median heuristic commits the entire ensemble to a single scalar bandwidth, our method optimizes an arbitrary number of continuous kernel parameters—for instance, one bandwidth per coordinate—and applies to any parameterized kernel family, affording a flexibility that a fixed rule cannot. Because the kernel update reuses the score evaluations $\nabla \log \pi$ already computed for the particle step, it adds no further evaluations of the target score, typically the dominant cost; the remaining overhead is controllable by the frequency and number of inner steps, and in our experiments was tuned to leave the per-iteration runtime essentially unchanged.
- (ii) **A convergence analysis of the adaptive scheme.** We analyze Ad-SVG in the discrete-time mean-field setting and lift the existing fixed-kernel convergence guarantees to the adaptive case. We prove that the maximal KSD over the parameterized kernel class converges to zero as the particle distribution approaches the target, both when the inner KSD maximization is solved exactly and when it is solved only approximately, the latter under an explicit asymptotic-optimality condition on the inner optimizer. Under an additional Stein logarithmic Sobolev inequality—which, notably, we require of only a *single* kernel in the class—we further obtain convergence in KL divergence together with explicit iteration-complexity bounds in the mean-field limit.
- (iii) **Empirical validation across dimensions and tasks.** Through numerical experiments spanning Gaussian mixtures, Gaussian targets of increasing dimension, an ODE-based linear inverse problem, Gaussian-process inference, and Bayesian logistic regression, we show that adaptive kernel selection matches the median heuristic on well-conditioned low-dimensional targets and consistently outperforms it as the dimension and anisotropy grow, resolving in particular the variance collapse that afflicts median-bandwidth SVGD.

2 Mathematical background

We fix throughout a symmetric positive-definite kernel $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ with associated RKHS $\mathcal{H}_0$, and we take $\mathcal{H}$ to be the d -fold Cartesian product of $\mathcal{H}_0$, endowed with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$ and canonical feature map $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$. We write $x \cdot y$ for the Euclidean inner product and $\nabla \cdot$ for the divergence operator. The space of probability measures on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is denoted $\mathcal{P}(\mathbb{R}^d)$, and $\mathcal{P}_p(\mathbb{R}^d)$ its subspace of measures with finite p -th moment. For

$\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, the Wasserstein p -distance is defined as

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ denotes the set of couplings of μ and ν , that is, the probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ whose marginals are μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [**IPM**] quantify the distance between two measures as the maximal discrepancy of their integrals over a prescribed class of test functions. Rendering this construction tractable for the distance between a sample and an intractable target requires a class of test functions with zero mean under the target; Stein’s method [**bounderrornormal**] furnishes precisely such a class. Concretely, for the Stein operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and a suitable kernel, one has $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for all $f \in \mathcal{H}$ [**KernelTestGoodness, KernelizedSteinDiscrepancy**]. Adopting the RKHS unit ball as the test-function class then yields the *kernelized Stein discrepancy* (KSD),

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the unit ball in $\mathcal{H}$. The maximizer admits the closed form $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$ with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, where $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$ and Φ_k is the feature map of the kernel k . Substituting f^* collapses the supremum onto the RKHS norm of ψ , and delivers the equivalent, computationally convenient characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target distribution π and a reference distribution μ_0 , SVGD transforms μ_0 into an approximation of π through the sequence of pushforwards $\mu_{n+1} := T_{\sharp} \mu_n$, in which $T_{\sharp}$ is the pushforward operator associated with the transport map $T(x) = x + \gamma \psi^{\mu_n}(x)$ and the vector field $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, $\mu \in \mathcal{P}(\mathbb{R}^d)$, is the direction of steepest descent of the KL divergence. This choice is far from arbitrary: the closed-form solution of Equation (1) shows that $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ is exactly the minimizer of $\frac{d}{d\gamma} \text{KL}(T_{\sharp} \mu_n \| \pi) \Big|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [**SVGD**], with $\text{KL}(\cdot \| \cdot)$ the KL divergence. Consequently, along this direction the instantaneous decrease of the KL divergence coincides with the squared KSD,

$$\frac{d}{d\gamma} \text{KL} \left((\text{Id} + \gamma \psi^{\mu_n})_{\sharp} \mu_n \parallel \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n|\pi). \quad (2)$$

Applying the maps T so constructed iteratively to a particle set $\{X_0^i\}_{i=1}^M$ drawn from μ_0 , and approximating the vector field by its empirical counterpart over the particles, yields the SVGD particle updates

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because establishing convergence of SVGD directly in the KL divergence is difficult and demands restrictive assumptions, recent analyses have instead measured progress through the kernelized Stein discrepancy (KSD) [NEURIPS20203202111c, NEURIPS2023'54e5d7af, pmlr-v162-salim22a]. This shift, however, brings to the surface a subtlety that motivates our approach: driving the KSD to zero does not on its own guarantee weak convergence, because the sequence of mean-field measures may fail to be tight—a necessary condition for weak convergence of probability measures on Polish spaces. Moreover, as the variational characterization in Equation (1) makes explicit, it is precisely the RKHS structure, and hence the kernel, that determines whether convergence in KSD upgrades to weak convergence [pmlr-v70-gorham17a]. The kernel therefore governs not only the speed of the dynamics but the very meaning of the convergence they attain.

To exploit this dependence rather than merely endure it, we formulate our adaptive variant of SVGD over a parameterized family of kernels $\{k_\theta \mid \theta \in \Theta\}$. For each $\theta \in \Theta$ we write KSD_θ for the corresponding *kernelized Stein discrepancy*, ψ_θ^μ for the induced optimal update direction, and Φ_θ for the associated feature map. Rendering the parameterization explicit allows the kernel to be selected adaptively during optimization, so that its influence on convergence becomes a degree of freedom to be optimized rather than a fixed choice to be tolerated.

The guiding principle is to select, at each iteration, the kernel that maximizes the KSD between the current empirical particle distribution and the target. The justification is immediate: as Equations (2) and (7) show, the instantaneous decrease of the KL divergence under an SVGD step equals the squared KSD at the current particle measure, so the kernel attaining the largest KSD is exactly the kernel that maximizes the rate of KL decrease at that iteration. The gradient-flow geometry of SVGD reinforces this reasoning: kernels that induce larger KSD values correspond to RKHSs with more favorable convergence properties [Nskn2021SteinVG, Duncan2019OnTG]. Maximizing the KSD over Θ therefore improves both the per-step progress and the geometry of the flow simultaneously.

Whereas vanilla SVGD aims only to drive the KSD to zero for a single, fixed kernel, our algorithm optimizes over the entire class, targeting at each iteration the largest attainable KSD. Unlike the median heuristic of **SVGD**, the mechanism is tied to no particular kernel form but applies to any parameterized family: at each step we seek an optimal parameter

$$\theta_n \in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_\theta(\mu_n | \pi).$$

We realize this by taking one or more gradient-ascent steps on the KSD with step size $s > 0$ before performing the particle update; to control runtime, the parameter may be refreshed only intermittently rather than at every SVGD step (see Algorithm 1), a choice governed by a user-specified decision variable `paramupdate`. The design is deliberately modular. The base transport step may be replaced by more advanced SVGD variants—adaptive step-size schedules, line searches, momentum, or second-order and preconditioned updates—without disturbing the kernel-selection mechanism, and the gradient ascent on θ may equally be exchanged for any alternative optimizer. Our concrete implementation of the kernel update rests on the closed-form representation of the squared KSD [**KernelTestGoodness**, **KernelizedSteinDiscrepancy**],

$$\text{KSD}^2(\mu | \pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k \, d(\mu \otimes \mu) \quad (4)$$

with

$$u_{\pi}^k(x, y) = k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \quad (5)$$

from which the gradient with respect to θ follows by direct differentiation. Two observations bound the resulting overhead relative to vanilla SVGD.

- The gradient-ascent steps for the kernel parameter reuse exactly the score evaluations $\nabla \log \pi(X_n^i)$ already required by the corresponding SVGD step. Ad-SVGD therefore incurs *no* additional evaluations of $\nabla \log \pi$, which are typically the dominant cost; the extra work is confined to the kernel-derivative computations of Equations (4) and (5).
- The remaining cost is controllable by design. It can be curtailed by taking only a small number of inner update steps nstepstheata for the kernel parameter, by refreshing the parameter only once every several SVGD steps, or, when the number of particles M is large, by estimating the KSD from a subsample of the particles. In our experiments these controls were set so that the per-iteration runtime of Ad-SVGD was essentially indistinguishable from that of median-heuristic SVGD; a systematic characterization of the cost as the parameter dimension and particle count scale is left to future work.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_{\theta} \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps $\text{nsteps}, \text{nstepstheata} \in \mathbb{N}$
Output: Final particle set $\{X_{\text{nsteps}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

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for  $n = 0$  to  $\text{nsteps} - 1$  do
    Decide  $\text{paramupdate} \in \{\text{True}, \text{False}\}$ ;
    if  $\text{paramupdate}$  then
         $\theta_n^0 \leftarrow \theta_{n-1}$ ;
        for  $\ell = 0$  to  $\text{nstepstheata} - 1$  do
             $\theta_n^{\ell+1} \leftarrow \theta_n^{\ell} + s \nabla_{\theta_n^{\ell}} \text{KSD}_{\theta_n^{\ell}}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
        end
         $\theta_n \leftarrow \theta_n^{\text{nstepstheata}}$ ;
    else
         $\theta_n \leftarrow \theta_{n-1}$ ;
    end
    for  $i = 1$  to  $M$  do
         $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
    end
end

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4 Convergence analysis

We consider a target measure with Lebesgue density $\pi(x) \propto \exp(-V)$ determined by a potential $V : \mathbb{R}^d \rightarrow \mathbb{R}$. To place Ad-SVGD on a firm theoretical footing, we extend recent KSD-based

convergence results for SVGD to the adaptive setting. We build specifically on the analysis of [pmlr-v162-salim22a], which operates under Talagrand’s inequality—a condition that holds under mild assumptions on the target and is strictly weaker than the logarithmic Sobolev inequality commonly invoked elsewhere. We adopt the same assumptions as [pmlr-v162-salim22a], now required to hold uniformly over all kernel parameters $\theta \in \Theta$; the uniform boundedness of Assumption 3 is the price of ranging over the whole kernel class rather than a single fixed kernel.

Assumption 1. *We assume that $V \in C^2$ such that $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$ and that the Hessian H of V is uniformly bounded w.r.t. the operator norm, i.e. there exists $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for all $x \in \mathbb{R}^d$.*

Assumption 2. *We assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s inequality T1, which means that there exists $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \| \pi) / \lambda}$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. *We assume there exists $B > 0$ such that $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$, and that $\nabla \Phi_\theta$ is continuous with $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$, $\theta \in \Theta$.*

4.1 Convergence analysis of SVGD under Talagrand’s inequality

In the following, we denote

$$\mathcal{F}(\mu) := \text{KL}(\mu \| \pi).$$

and rely on the following fundamental descent inequality. Fix a kernel parameter $\theta \in \Theta$ for which Assumption 3 holds, and, for an arbitrary direction $g \in \mathcal{H}$, define the pushforward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu.$$

Under Assumptions 1 and 3, and for $\gamma, B > 0$, $\alpha > 1$, and $g \in \mathcal{H}_\theta$ subject to the step-size restriction $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, Proposition 3.1 in [pmlr-v162-salim22a] gives

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta}^2 \quad (6)$$

with $K = (\alpha^2 + L)B$. Applied to the iterates (3) with a fixed kernel parameter $\theta \in \Theta$, this inequality yields the descent condition [pmlr-v162-salim22a],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\|) + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1}. \quad (8)$$

The crux of (7) lies in verifying the step-size restriction $\gamma \|\psi_\theta^{\mu_n}\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$, which follows from Talagrand’s inequality (Assumption 2) and licenses the application of (6) with $g = \psi_\theta^{\mu_n}$. Summing

(7) over the iterates then telescopes to $\sum_{n=0}^{\infty} \text{KSD}_{\theta}^2(\mu_n|\pi) \leq c_{\gamma}^{-1} \mathcal{F}(\mu_0)$ with $c_{\gamma} = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$, whence the vanishing of the KSD,

$$\lim_{n \rightarrow \infty} \text{KSD}_{\theta}^2(\mu_n|\pi) = 0,$$

follows immediately.

Introducing an adaptive kernel-parameter sequence (θ_n) replaces the fixed θ in the descent condition (7) by the running parameter θ_n :

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) \text{KSD}_{\theta_n}^2(\mu_n|\pi).$$

4.2 Convergence analysis of Ad-SVGd.

We treat first the idealized case in which the inner KSD maximization is solved exactly, so that the adaptive iteration reduces to the simplified form

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi), \end{aligned} \tag{9}$$

so that

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n|\pi).$$

This exact formulation is available whenever the inner KSD maximization admits a (unique) closed-form solution. Such is the case, for instance, in the multiple-kernel SVGd framework of [ai'stein'2023], which parameterizes the kernel as a convex combination of base kernels,

$$k_{\theta}(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1,$$

and which we recover here as a special case of our general construction. In the general setting the maximizer θ_n in (9) must instead be approximated, and we assume throughout that $\operatorname{argmax}_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The following descent condition is an immediate consequence of Theorem 3.2 in [pmlr-v162-salim22a]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

there exists $c_{\gamma} > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) = 0$.*

We now turn to the practically relevant case in which the inner maximization over the kernel parameter is solved only approximately, as it is by the gradient ascent in Algorithm 1. Assuming $\max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu|\pi) < \infty$ for every $\mu \in \mathcal{P}(\mathbb{R}^d)$, we study the alternating scheme

$$\begin{aligned}\theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\text{Id} + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n\end{aligned}\tag{10}$$

for a sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ designed to increase $\text{KSD}_{\theta}(\mu_n | \pi)$. In Algorithm 1, Ψ_n is the gradient-ascent update on the KSD, although, as noted above, any other suitable iterative optimizer could be substituted without affecting the analysis. The sole property we require of the update rule is the following asymptotic optimality.

Assumption 4. *We assume that there exists a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ such that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and*

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

This condition asks that the inner optimizer close the KSD suboptimality gap asymptotically; it is met, in particular, whenever the gradient ascent attains a global maximizer in the limit. Under it we obtain the following convergence guarantee for the approximate scheme.

Theorem 3. *Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Using Theorem 3.2 in [pmlr-v162-salim22a] we obtain

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

Since $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is decreasing and bounded below by zero, it converges and, in particular, it is a Cauchy sequence. Hence, we have

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Under Assumption 4 we obtain

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under an additional Stein logarithmic Sobolev inequality, this guarantee strengthens from convergence of the KSD to convergence in the KL divergence itself, and, for specific decay profiles of ε_n , it yields explicit iteration-complexity bounds for the mean-field limiting system (10). The same setting further affords an explicit characterization of the asymptotic bias of Ad-SVGD when the inner maximization is solved only to a fixed $\bar{\varepsilon}$ -accuracy at each iteration; the full statements and proofs are deferred to Appendix B. A notable feature of the adaptive formulation is that the Stein log-Sobolev inequality need hold for only a *single* kernel in the class $\{k_{\theta}\}$, a requirement considerably easier to verify than the fixed-kernel counterpart, in which the chosen kernel must satisfy it outright.

5 Numerical Experiments

We now evaluate Ad-SVGD empirically. The main text reports two representative experiments—a one-dimensional Gaussian mixture and a linear inverse problem governed by an ordinary differential equation—together with a Bayesian logistic regression study on the Covertype data set (Section 5.4) that follows the evaluation protocol established in the prior SVGD literature. To isolate the effect of dimension, Appendix A supplements these with a systematic scaling study on multivariate Gaussian and Gaussian-process targets of growing dimension. Throughout, we take median-heuristic SVGD—the standard and most widely deployed instantiation of the method—as the baseline against which adaptive kernel selection is measured. Across all of these settings a single conclusion emerges consistently: Ad-SVGD matches the heuristic on well-conditioned low-dimensional targets and decisively outperforms it as the dimension grows, chiefly by correcting the systematic underestimation of posterior variance to which the heuristic is prone.

5.1 Kernel parameterization

SVGD is most commonly instantiated with kernels of the form

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ is the p -norm on $\mathbb{R}^d$ [SVGD, DBLP:conf/iclr/BaEGSSWZ22, Duncan2019OnTG]. Our study centers on the choice of the scalar parameter h , the kernel *bandwidth*. The standard heuristic sets $h = \frac{\text{med}^p}{\log(M-1)}$, with med the current median pairwise distance between particles, a rule motivated by the aim of balancing the two terms of the SVGD update (3) [SVGD]. Its defining weakness is that it binds the entire ensemble to a single bandwidth shared across all coordinates, and so cannot accommodate targets whose scales differ from one direction to the next.

Ad-SVGD lifts exactly this restriction. To exploit its flexibility, we employ anisotropic product kernels

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

parameterized by a vector $h = (h_1, \dots, h_d)$ of coordinate-wise bandwidths, so that each dimension is endowed with its own length scale. The derivatives required to apply Algorithm 1 to these kernels—through Equations (4) and (5)—are available in closed form and inexpensive to evaluate. As a control, we also tested a naive per-dimension analogue of the median heuristic on the same kernel family; this variant performed poorly and, tellingly, again succumbed to variance collapse. The contrast is instructive: it confirms that per-coordinate bandwidths confer no benefit in themselves, and that the gains we report stem specifically from choosing them adaptively rather than by a fixed rule. Because these product kernels assign a length scale per coordinate, they are best suited to targets whose principal directions of variation are approximately axis-aligned; strongly rotated covariance structures would call for the richer, non-diagonal parameterizations discussed next.

The construction extends well beyond the kernels used here. Because the adaptive mechanism is indifferent to the specific kernel family, it accommodates richer and more scalable parameterizations for higher-dimensional or non-axis-aligned targets: spectral or low-rank parameterizations can track the dominant directions of variability—including off-diagonal correlation structure that coordinate-wise product kernels cannot represent—without the cost of a full parameterization, while Matérn

or mixed-product constructions afford additional control over smoothness and anisotropy. We restrict the experiments below to diagonal product kernels in order to isolate the effect of adaptive bandwidth selection in its simplest form; exploiting these richer families is a natural direction for extending the method to strongly correlated posteriors.

5.2 Toy example

We begin with the one-dimensional example of **SVGD**, the two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. Initializing the particles from $\mathcal{N}(0, 1)$ with $M \in \{50, 200, 500\}$, we run SVGD for 10^4 steps at step size 1, using the kernel above with $p = 1$ across a range of fixed bandwidths h . Sample quality is measured by the Wasserstein 1-distance $\mathcal{W}_1$, computed from the explicit one-dimensional formula $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_{\mu}(x) - F_{\nu}(x)| dx$ [Statistical Aspects of Wasserstein Distances] against an exact reference sample of size 10^5 . As Figure 1 makes plain, accurate approximation is attained only for bandwidths within a narrow range, with the error rising sharply on either side. This pronounced sensitivity of the final kernel dynamics to h is exactly the failure mode that a principled selection of bandwidth must address, and that is the point of departure for Ad-SVG.

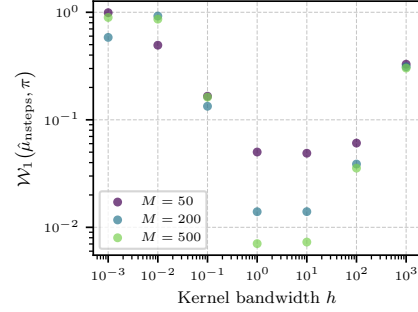


Figure 1: Final Wasserstein 1-distances for the one-dimensional example using SVGD with a range of fixed bandwidths h ($M \in \{50, 200, 500\}$ particles). Accuracy is high only within a narrow band of h , and the sensitivity of the final kernel dynamics to h is exactly the failure mode that a principled selection of bandwidth must address, and that motivates adaptive selection.

5.3 Linear inverse problem based on ODE

Our second experiment, adapted from [Weissmann2022Optimization], moves to a genuinely multidimensional posterior arising from a functional inverse problem, a setting in which anisotropy is intrinsic rather than contrived. We consider the one-dimensional differential equation

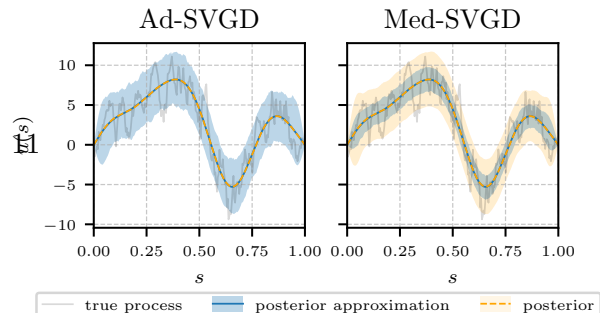
$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

together with the associated inverse problem of recovering the right-hand side $u(\cdot) \in L^2([0, 1])$ from finitely many noisy observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation model reads

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ denotes observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ factors as $\Phi = \mathcal{O} \circ H^{-1}$, with solution operator $H(f) = -f'' + f$ and observation operator $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^\top \in \mathbb{R}^{N_{\text{obs}}}$ evaluating the solution at the N_{obs} equidistant points $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation, we endow u with a Gaussian process (GP)



prior specified through the truncated Karhunen–Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

with basis functions $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and independent coefficients $x_k \sim \mathcal{N}(0, \lambda_k)$, $\lambda_k = 50k^{-2}$. Inference thus reduces to estimating the vector of KL coefficients $x = (x_1, \dots, x_{N_x})^\top \in \mathbb{R}^{N_x}$, endowed with the Gaussian prior $\mathcal{N}(0, \Gamma_0)$, $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Under the Gaussian noise model $\varepsilon \sim \mathcal{N}(0, \Gamma)$, the posterior density is

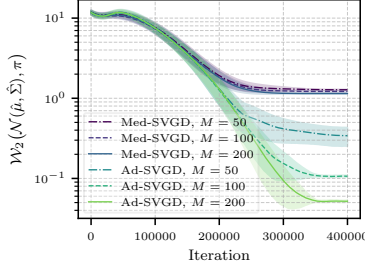
$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

In the implementation we discretize Equation (11) on a grid of mesh size 2^{-8} , replacing H by the corresponding numerical solution operator, and consider the fully observed system $N_{\text{obs}} = 2^8$. We truncate the KL expansion of u at $N_x = 16$ terms and set the noise covariance to $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$. We compare SVGD under the median heuristic (denoted Med-SVGD) against Ad-SVGD across ensemble sizes $M \in \{50, 100, 200\}$, using the product kernels of Section 5.1 with $p = 1$. Both methods are run for $4 \cdot 10^5$ iterations at particle step size 10^{-3} with the AdaGrad-style adaptive step-size control recommended by **SVGD**. In Ad-SVGD the bandwidths are updated with step size 10^{-5} and only once per 100 particle updates; under this schedule the two methods showed no appreciable runtime difference, so that the comparison isolates the effect of kernel adaptation at essentially equal computational cost.

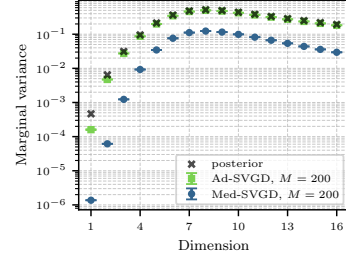
Figure 2 shows the GP reconstruction for a representative seed. Both methods recover the posterior mean accurately, yet only Ad-SVGD reproduces the posterior uncertainty; Med-SVGD collapses it. To quantify approximation quality we adopt the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ formed from the particle sample mean $\hat{\mu}$ and sample covariance $\hat{\Sigma}$ [**StatisticalAspectsWasserstein**]. Since π is itself multivariate normal, this distance is available in closed form,

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π denote the posterior mean and covariance. We additionally compare the marginal variances of the final particle distribution with those of the posterior. Figure 3 aggregates these results over 56 random seeds (the posterior covariance being independent of $\bar{y}$). The pattern is unambiguous. Ad-SVGD approximates the posterior markedly more accurately than Med-SVGD, which systematically underestimates the uncertainty; and, tellingly, only Ad-SVGD benefits from additional particles, its approximation quality continuing to improve as M grows beyond 50 while that of Med-SVGD stagnates. Figure 4 traces the bandwidths selected by Ad-SVGD. The component-wise bandwidths stabilize well before the approximation error does and separate clearly across components, their final values correlating negatively with the corresponding marginal variances—direct evidence that the method automatically assigns finer length scales to the more concentrated directions of the posterior.

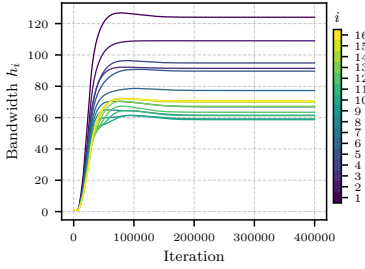


(a) Convergence of the Wasserstein-2 approximation error to the posterior: Ad-SVGD attains a lower error and continues to improve as M grows.

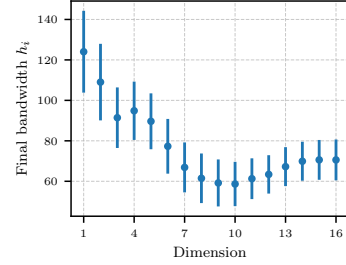


(b) Marginal variances of the final particles: Ad-SVGD recovers the posterior variances that Med-SVGD underestimates.

Figure 3: ODE-based inverse problem: comparison of Med-SVGD and Ad-SVGD (mean and 95% confidence interval over 56 random seeds). Across both metrics Ad-SVGD approximates the posterior more accurately and, unlike Med-SVGD, faithfully recovers the posterior uncertainty.



(a) Evolution of the per-dimension bandwidths (mean), which stabilize well before the approximation error does.



(b) Final bandwidths (mean and 95% confidence interval), clearly differentiated across components.

Figure 4: Behavior of the coordinate-wise bandwidths selected by Ad-SVGD for the ODE-based inverse problem, aggregated over 56 random seeds. The bandwidths adapt automatically to the anisotropy of the posterior, assigning distinct length scales to distinct components.

5.4 Bayesian logistic regression

To assess Ad-SVGD on a standard real-data benchmark, we follow **SVGD** and **pmlr-v151-liu22a** in studying a Bayesian logistic regression (BLR) model on the Covtype data set [**covtype**]. The regression weights w receive a hierarchical Gaussian prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with $\alpha \sim \text{Gamma}(1, 0.01)$, and inference targets the joint posterior of $x = [w, \alpha]$. Following **pmlr-v151-liu22a**, we obtain a high-quality posterior reference from the No-U-Turn Sampler (NUTS) of **nuts**, against which the approximation quality of both Med-SVGD and Ad-SVGD is measured. Following **pmlr-v151-liu22a**, we evaluate the methods on subsets of size 1000 and aggregate results over 10 random draws. Predictive performance on the classification task is quantified by the test-set accuracy of the particle-mean classifier. As Figure 5 shows, Ad-SVGD outperforms Med-SVGD on nearly every seed, attaining both higher test accuracy and closer agreement with the MCMC reference; adaptive kernel selection thus improves the downstream predictive task and the posterior fit in tandem.

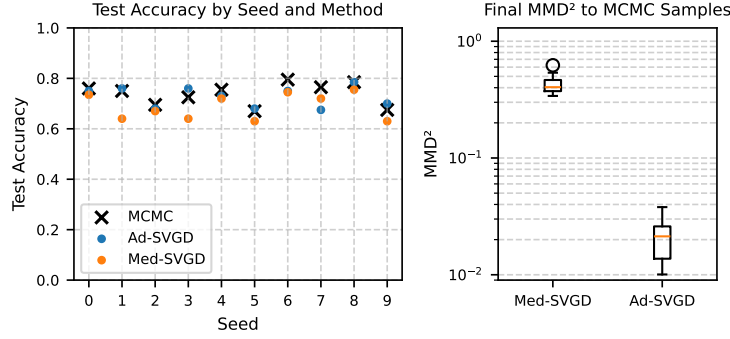


Figure 5: Approximation quality of the SVGD particles for Bayesian logistic regression, measured by test-set prediction accuracy (left) and squared MMD to the NUTS reference samples (right). Ad-SVGD improves predictive accuracy while at the same time moving closer to the MCMC reference.

To gauge the fidelity of the posterior approximation itself, the same figure reports the squared maximum mean discrepancy (MMD^2) [**MMDref**] to the MCMC samples, on which the two methods differ substantially. This gap is corroborated by Figure 7, which compares the covariance matrices of the final particles with the MCMC reference, averaged over the 10 seeds and rendered in the visualization style of **pmlr-v151-liu22a**: Med-SVGD severely underestimates the posterior uncertainty, whereas Ad-SVGD reproduces it far more faithfully. Finally, Figure 6 shows that, starting from the entirely uninformed initialization $h_i = 1$ in every component, Ad-SVGD autonomously recovers component-wise bandwidths spanning several orders of magnitude—confirming that the adaptation is driven by the data rather than by the initialization.

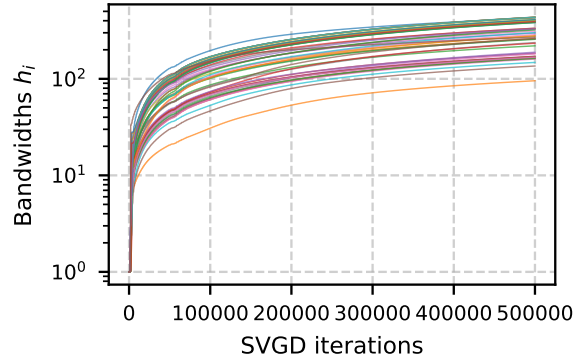


Figure 6: Evolution of the coordinate-wise bandwidths for the BLR example. From the uninformed initialization $h_i = 1$, Ad-SVGD recovers bandwidths that differ across components by several orders of magnitude—confirming that the adaptation is driven by the data rather than by the initialization.

6 Limitations and outlook

The principal limitation of our analysis is its reliance on Assumption 4, which posits that the inner optimizer drives the KSD suboptimality ε_n to zero. Since the KSD is in general a nonconvex function of the bandwidth parameters, establishing when the alternating gradient-ascent scheme of Algorithm 1 provably satisfies this condition—and, more broadly, characterizing the local maxima the inner ascent may reach—remains open and is the subject of ongoing work. We do not yet furnish such a guarantee; what we can report is that the scheme meets the assumption in practice across every setting we study, converging reliably from uninformed bandwidth initializations without manifesting the degeneracies, such as vanishing bandwidths or unstable updates, that unconstrained

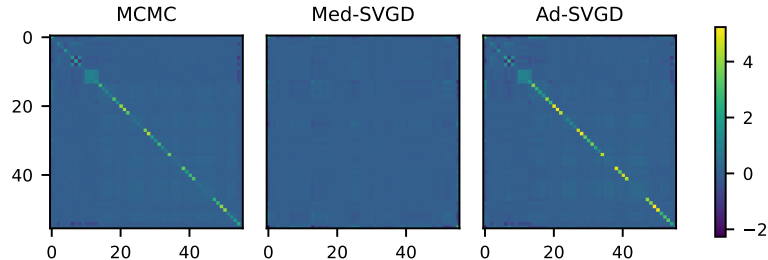


Figure 7: Covariance matrices of the final SVGD particle sets compared with the MCMC reference (mean over 10 seeds). Med-SVG collapses the posterior covariance, whereas Ad-SVG closely reproduces the reference structure.

KSD maximization might in principle admit. A complementary limitation is representational: the diagonal product kernels used in our experiments assign one length scale per coordinate and therefore capture axis-aligned anisotropy rather than strongly rotated covariance structure. The adaptive mechanism itself is agnostic to this choice, and the spectral, low-rank, and mixed-product families noted in Section 5.1 offer a direct route to non-diagonal structure; a systematic study of such kernels, together with a careful accounting of wall-clock cost, memory, and sensitivity to the kernel-update schedule as the parameter dimension and particle count grow, we leave to future work.

Our analysis and experiments deliberately target the original SVGD dynamics, so as to isolate the effect of kernel adaptation from the many other modifications that have been proposed. Precisely because the adaptive mechanism is modular and leaves the transport step untouched, a natural next step is to compose it with recent structural variants—sliced SVGD [gong2021sliced], Grassmann SVGD [pmlr-v151-liu22a], and Stein transport [nsken2024steintransportbayesianinference]—and with the matrix-valued and preconditioned schemes that likewise address anisotropy, where the two forms of improvement should prove complementary. Extending the evaluation to genuinely large-scale posteriors, such as Bayesian neural networks, is a further avenue that our present study, focused on moderate dimensions, does not yet address.

A More numerical experiments

A.1 Gaussian mixture models

We return to the example of Section 5.2 to examine it more closely. For the setup described there, Figure 9 displays histograms of the final particle distribution against the target density for the three fixed bandwidths $h \in \{0.001, 1, 1000\}$, illustrating how strongly the fixed-kernel outcome hinges on h . We then compare Med-SVGD with Ad-SVGD, contrasting the final particle distributions obtained after 10^4 iterations at step size 1 across a range of particle counts M .

Figure 8 reports the Wasserstein 1-distance between the final empirical particles and the target for $M = 10, 20, 50, 100, 200, 500$. As anticipated, accuracy improves with dimensionality for both methods, which remain essentially indistinguishable, each reaching a Wasserstein distance below 0.01 at $M = 500$. In this benign one-dimensional regime, adaptivity is therefore not required to match the heuristic; the advantage of Ad-SVGD emerges, as we show next, precisely where the heuristic breaks down.

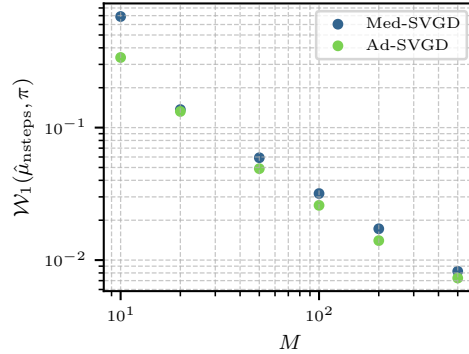


Figure 8: Approximation error (W_1) for the 1D Gaussian mixture as a function of the particle count M . Both

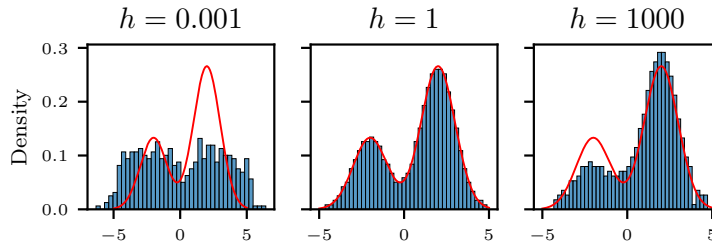


Figure 9: Histograms of the final particle set for fixed bandwidths $h = 0.001, 1, 1000$, with the target density π overlaid in red. Only the intermediate bandwidth recovers the bimodal target, while both extremes fail—a visualization of the sensitivity quantified in Figure 1.

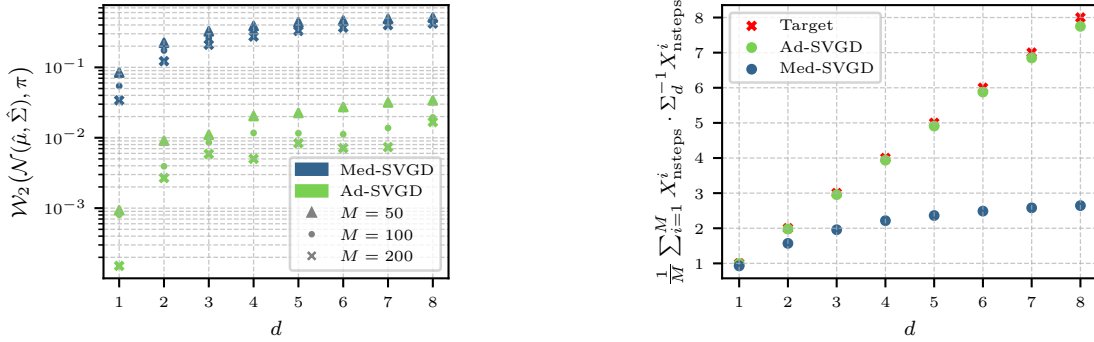
In the one-dimensional case, then, both methods approximate the target well, and adaptive bandwidth selection performs on par with the median heuristic without conferring a decisive advantage. This is the expected outcome for a low-dimensional, well-conditioned target; it establishes the baseline against which the higher-dimensional experiments that follow stand out, where the heuristic degrades and adaptivity becomes essential.

A.2 Scaling dimension: multivariate normal distribution

To probe the effect of dimension directly, we consider the family of Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$, an axis-aligned yet increasingly ill-conditioned

sequence whose marginal scales span an order of magnitude at $d = 8$. The particles are initialized from $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$ and the algorithms are run for 10^4 iterations at step size 0.1, with the step size reduced and the iteration count increased correspondingly whenever numerical stability required it.

We track approximation quality as the dimension grows. Since the Wasserstein 2-distance between two Gaussians admits an explicit formula [**StatisticalAspectsWasserstein**], we adopt as our quality measure the Wasserstein 2-distance between π_d and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ built from the empirical particle mean $\hat{\mu}$ and covariance $\hat{\Sigma}$ [**StatisticalAspectsWasserstein**]; following the Python Optimal Transport library [**PythonOptimalTransport**] used for the computation, we refer to it as the *Bures–Wasserstein distance*. Figure 10a plots this quantity for Med-SVGD and Ad-SVGD with $M = 50, 100, 200$ as a function of d . Ad-SVGD outperforms Med-SVGD at every dimension, and by a margin that widens with d , indicating that its advantage grows in step with dimensionality across the range examined.



(a) Bures–Wasserstein approximation error for $M \in \{50, 100, 200\}$: Ad-SVGD dominates Med-SVGD at every dimension, by a widening margin.

(b) χ^2 -test statistic for $M = 200$: Ad-SVGD tracks the true value d , whereas Med-SVGD drifts steadily away as d grows.

Figure 10: Multivariate normal targets of increasing dimension. Both an aggregate distance (left) and a variance-sensitive statistic (right) show that Ad-SVGD scales gracefully with dimension while Med-SVGD deteriorates.

Because the Bures–Wasserstein distance is relatively insensitive to the small variances in the trailing components of π_d , we complement it with a statistic that exposes any failure to capture that uncertainty directly: the quadratic form $X \cdot \Sigma_d^{-1} X$, which for $X \sim \mathcal{N}(0, \Sigma_d)$ is χ^2 -distributed with d degrees of freedom and hence has expected value d (cf. [**Metropolisadjustedinteracting**]). Figure 10b compares the particle-averaged value of this statistic ($M = 200$) with its true expectation d . The Med-SVGD statistic departs ever further from d as the dimension increases, whereas the Ad-SVGD statistic stays close to the true expectation throughout—a clear indication that Med-SVGD ceases to approximate the higher-dimensional targets faithfully while Ad-SVGD continues to do so.

Tables 1 and 2 furnish a finer, component-wise account for $M = 200$ across d , listing the marginal variances of the particle set alongside those of the target (top row) for Med-SVGD and Ad-SVGD, respectively. Ad-SVGD reproduces the target marginal variances accurately in every component and at every dimension, whereas Med-SVGD substantially underestimates the uncertainty—consistent with the χ^2 diagnostic of Figure 10b.

Table 1: Marginal variances of the final Med-SVGD particle distribution ($M = 200$) for the d -dimensional Gaussian targets, with the target marginal variances of π_d in the top row. Med-SVGD underestimates every marginal variance, and the deficit deepens as d grows.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Marginal variances of the final Ad-SVGD particle distribution ($M = 200$) for the d -dimensional Gaussian targets, with the target marginal variances of π_d in the top row. Ad-SVGD matches the target variances closely in every component and at every dimension.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

Finally, we examine the marginal particle distributions at $M = 200$ and $d = 8$ in full. For visualization we normalize the particles by the target marginal standard deviations (scaling the i -th component by i), which renders each marginal of π_d standard normal. Figures 11 and 12 display the resulting histograms for Med-SVGD and Ad-SVGD, each with a standard Gaussian density overlaid, and Figures 13 and 14 present the corresponding Q-Q plots against the standard normal. The two views converge on the same conclusion: Ad-SVGD captures every marginal accurately, while Med-SVGD systematically underestimates the spread of the target.

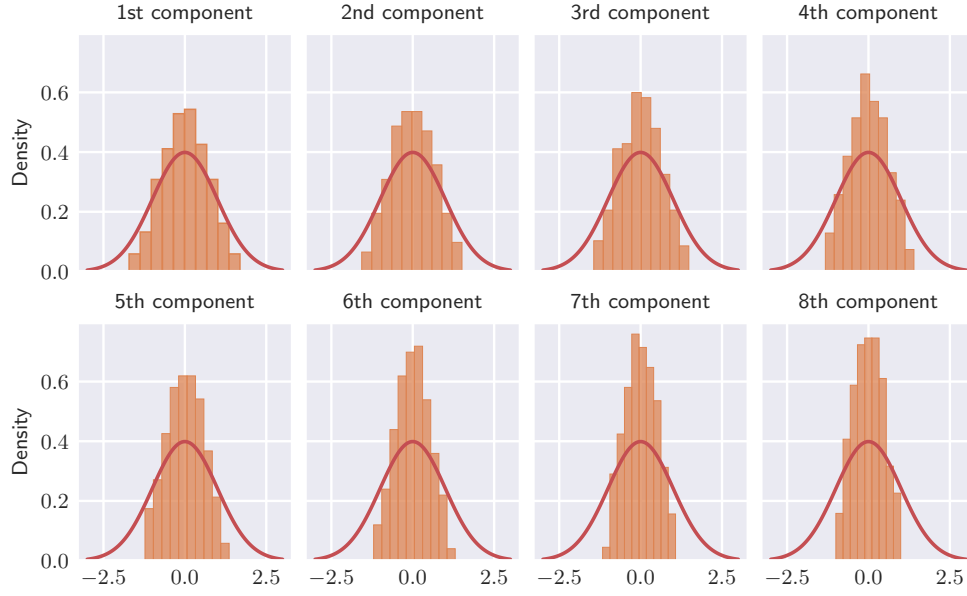


Figure 11: Component-wise histograms of the final Med-SVGD particles for the eight-dimensional example ($M = 200$), rescaled by the target marginal standard deviations so that each target marginal is standard normal (overlaid). The histograms are visibly narrower than the reference, exposing the variance underestimation of Med-SVGD.

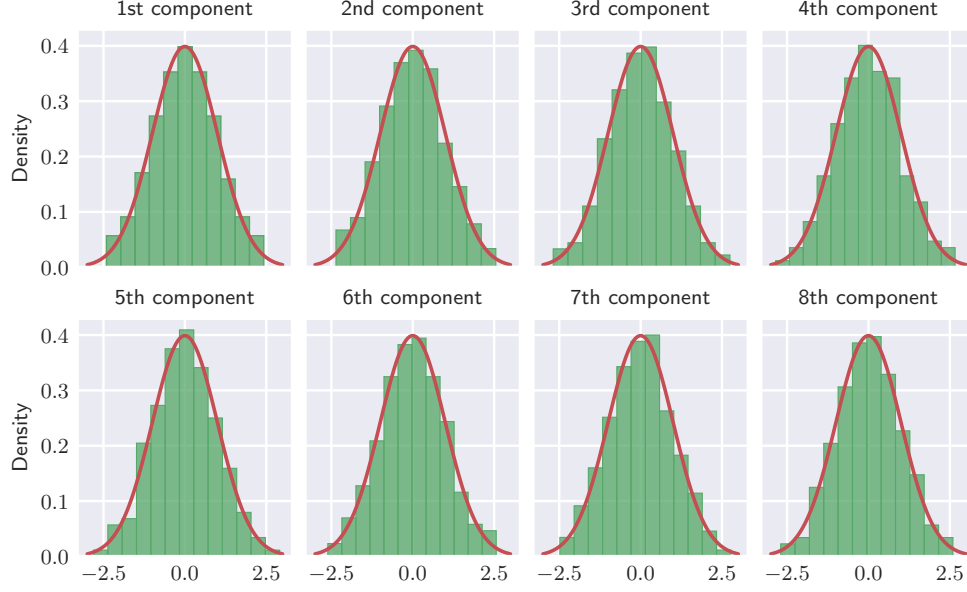


Figure 12: Component-wise histograms of the final Ad-SVGD particles for the eight-dimensional example ($M = 200$), rescaled by the target marginal standard deviations so that each target marginal is standard normal (overlaid). The histograms track the reference density closely across every component.

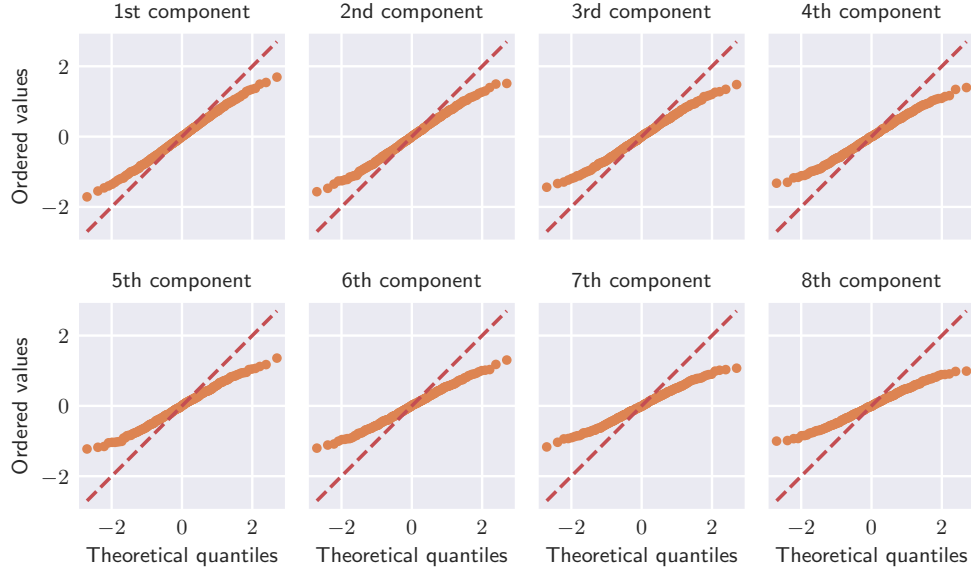


Figure 13: Q–Q plots of the rescaled marginals of the final Med-SVG particle distribution ($M = 200$, eight-dimensional example) against the standard normal, with the identity line ($y = x$) shown for reference. The systematic departure from the identity line reflects the underdispersion of Med-SVG across all components.

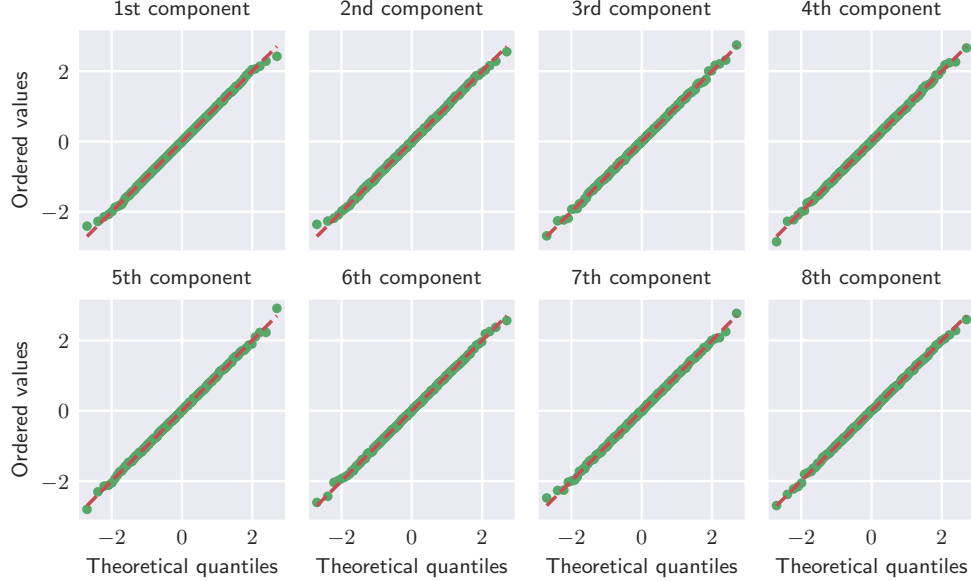


Figure 14: Q–Q plots of the rescaled marginals of the final Ad-SVG particle distribution ($M = 200$, eight-dimensional example) against the standard normal, with the identity line ($y = x$) shown for reference. The marginals align closely with the identity line, confirming the accurate recovery of the target marginals.

A.3 Gaussian process inference

As a further test of scalability, we study the Gaussian process (GP) inference task proposed in [reich21], in which the dimension can be varied through the truncation level of the representation—affording a controlled means of increasing the problem dimension while holding its structure fixed. We consider a GP on $[0, 1]$ represented by a truncated Karhunen–Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ collects independent Gaussian coefficients $x_k \sim \mathcal{N}(0, k^{-2})$. Observing the process at N_y equispaced points in $[0, 1]$ and inferring the coefficients x_k yields, for fixed N_x and N_y , a linear-Gaussian inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$, $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal Σ of entries k^{-2} , $k = 1, \dots, N_x$, and the noise is independent Gaussian, $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

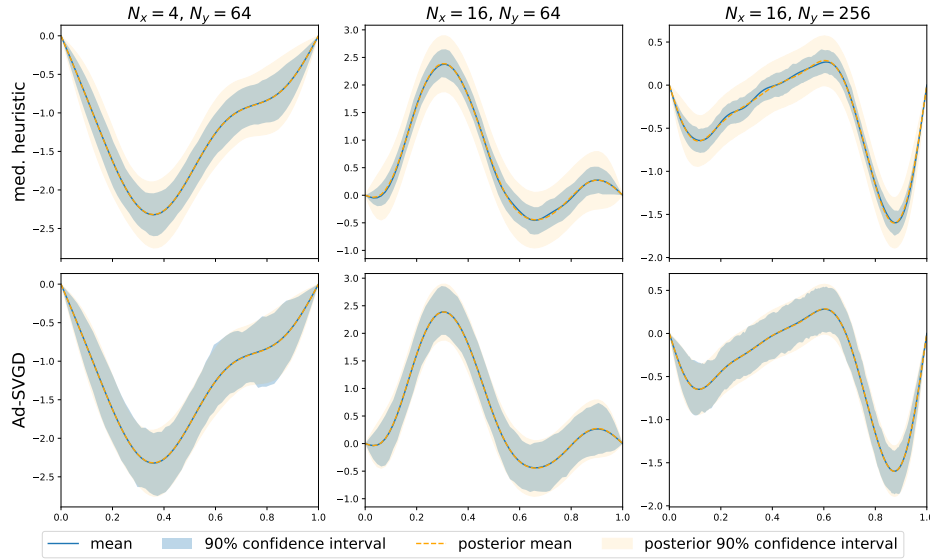


Figure 15: Estimated processes from SVGD with the median heuristic and from Ad-SVGD ($M = 100$ particles) for several model configurations, compared with the posterior. As the dimension grows, the median-heuristic band collapses whereas Ad-SVGD retains the posterior spread.

We apply SVGD to sample from the resulting posterior $X | Y = \bar{y}$ and again contrast the median heuristic with our adaptive approach. The median-heuristic runs employ the isotropic kernel $k(x, y) = \exp(-\|x - y\|_1/h)$, whereas Ad-SVGD uses the anisotropic product kernel $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with a distinct bandwidth h_i per dimension. As in [SVGD], an AdaGrad-style step-size control is used in the $N_x = 16$ configurations.

Figure 15 displays the estimated processes against the posterior for the three configurations $(N_x, N_y) \in \{(4, 64), (16, 64), (16, 256)\}$. As the dimension increases, the median heuristic underestimates the posterior variance ever more severely, whereas Ad-SVGD sustains an accurate approximation; because the contrast persists across the differing numbers of observations, the effect is attributable to dimension rather than to observation count. Table 3 renders the comparison quantitative through the trace of the particle covariance: the median heuristic recovers less than half of the total posterior variance in every configuration, while Ad-SVGD matches it closely throughout.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the covariance of the final particle distribution ($M = 100$ particles) compared with the theoretical posterior value for different (N_x, N_y) configurations, averaged over 25 runs. Ad-SVGD matches the theoretical trace closely in every configuration, whereas the median heuristic recovers less than half of it.

B Refined theoretical analysis

Our convergence result strengthens considerably under a Stein log-Sobolev inequality, which bounds the KL divergence directly by the KSD. This inequality was introduced by **Duncan2019OnTG**, who also examined the conditions under which it holds. A distinctive advantage of the adaptive setting is that the requirement may be relaxed substantially: it suffices that a *single* kernel in the entire class satisfy the log-Sobolev inequality, rather than every kernel individually. We formalize this relaxation as follows.

Assumption 5. *Given a kernel class $\{k_\theta \mid \theta \in \Theta\}$, we assume that $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \parallel \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

This property lets us quantify convergence directly in the KL divergence. Notably, it yields an explicit error bound on $\mathcal{F}(\mu_n)$ even under the merely approximate condition $\varepsilon_n \leq \bar{\varepsilon}$, without requiring $\varepsilon_n \rightarrow 0$: the algorithm then converges to a neighborhood of the target whose radius is controlled by the maximization accuracy $\bar{\varepsilon}$, so that better inner maximization directly buys a tighter asymptotic bias.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \parallel \pi) - \text{KSD}_{\theta_n}^2(\mu_n \parallel \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Iterating this recursion over $n \in \mathbb{N}$, we deduce the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We consider the three different cases separately:

1. Under $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by the computation of the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1 - \rho}.$$

2. Assuming that $\varepsilon_n \leq c_\varepsilon q^n$ for some $q \in (0, 1)$ and $c_\varepsilon > 0$ we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

Suppose that $q > \rho$, then we have

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1 - \rho/q} = q^n \frac{1}{q - \rho}$$

where we have used the formula for geometric series. Similarly, in the case of $q < \rho$, we can directly bound

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1 - q/\rho} = \rho^n \frac{1}{(\rho - q)}.$$

Finally, in the case $q = \rho$, we simply bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim line by line as before.

3. In the setting of $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we rewrite the upper bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned} \mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\ &= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\ &\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1 - \rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1 - \rho}. \end{aligned}$$

Here, we have used $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$ for all $k \leq \lfloor n/2 \rfloor$ and $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□